Čencov–Inversion Compatibility for CTMT Transport:
A Forced Admissibility Class and Tested-Class Structural Necessity

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Abstract

We construct a forced admissibility class for CTMT transport morphisms by intersecting two independent necessity principles: Čencov-style Fisher admissibility and inversion-stability of a regularized forward model. The goal is not to prove CTMT as a universal physical theory, but to define a morphism class that can survive mathematical review without obvious fatal objections and whose structural invariants survive a nontrivial synthetic adversarial battery.

The central claim is deliberately modest and tested-class: within the synthetic coil family previously studied, every scenario that satisfied Fisher admissibility, inversion stability, and dimensional closure preserved a common structural transport signature, while every violation scenario broke at least one component. Thus the structural signature acts as a necessary discriminator for the tested class, rather than a universal law. This refines earlier CTMT drafts by replacing exact equalities with stability bounds, using Tikhonov-regularized inversion, and making admissibility tolerance explicit.

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1 Introduction and aim

The present note has a narrower and more conservative aim than earlier CTMT drafts. It does not attempt to prove CTMT as a universal physical theory. Instead, it aims to:

> Define a forced admissibility class for CTMT transport morphisms, justified by independent necessity principles, and show that the structural invariants observed in the synthetic coil battery are necessary discriminators within the tested class.

In previous work on a synthetic coil battery, it was observed that Fisher monotonicity alone is insufficient to decide gauge equivalence and that gauge identification requires additional structural invariants, namely null-curvature stability and quotient-style Čencov stability. The coil results showed that radius scaling and compensated attacks can preserve Fisher monotonicity, winding, and self-reference while still failing null-curvature and quotient-class stability. This revealed a structural split between metric admissibility and transport-class structure.

The present note strengthens the mathematical side of that story. Instead of asserting strong equalities such as

$$\ker(J) = \ker(PJ), \qquad \mathrm{Im}(J) = \mathrm{Im}(PJ),$$

we work with Tikhonov-regularized inversion, admissibility tolerances, and stability bounds on projections and subspaces. The resulting statements are weaker but mathematically defensible and closer to modern inverse-problem practice.

The core message is:

- Čencov's theorem forces Fisher as the unique information metric under Markov morphisms.
- Inversion-stability of a regularized forward model forces approximate preservation of resolved and null-supported directions.
- Dimensional closure (mass balance, positivity, unit consistency) adds a physical admissibility layer.
- The intersection of these constraints defines a forced admissibility class.
- On the synthetic coil battery, the structural transport signature is a necessary discriminator for this tested class.

No universality beyond the tested class is claimed.

2 Fisher Geometry, Admissibility, and Čencov Background

We summarize the Fisher-information layer used throughout CTMT. This layer provides the distinguishability structure and the admissibility tolerance inherited from Čencov's theorem.

2.1 Fisher Geometry

Definition 1 (Fisher matrix)**.** Let $J(\theta) \in \mathbb{R}^{m \times d}$ be a local Jacobian of a forward model at parameter θ, and let C be a covariance matrix on the observation space. The local Fisher matrix is

$$F(\theta) = J(\theta)^\top C^{-1} J(\theta).$$

Remark 1. Čencov's theorem states that, up to a scalar factor, the Fisher metric is the unique Riemannian metric invariant under Markov morphisms. We adopt Fisher as the admissible information metric and require that admissible morphisms respect Fisher monotonicity up to tolerance.

2.2 Positive Minimum Eigenvalue

Definition 2 (Positive minimum eigenvalue)**.** For a symmetric positive semidefinite matrix $F \succeq 0$, define

$$\lambda^+_{\min}(F) = \min\{\lambda_i(F) : \lambda_i(F) > 0\}.$$

This quantity sets the natural scale for admissibility tolerance.

2.3 Fisher Admissibility Tolerance

Definition 3 (Fisher admissibility)**.** Let $\alpha > 0$ be a tolerance parameter. A coarse-grained Fisher matrix F_{cg} is Fisher-admissible if

$$F_{\mathrm{cg}} \preceq F + \varepsilon I, \qquad \varepsilon = \alpha\, \lambda^+_{\min}(F).$$

This tolerance-based definition avoids brittle exact-ordering requirements and matches the numerical practice used in the synthetic coil battery.

2.4 Observable Equivalence

Definition 4 (Observable equivalence)**.** Two Fisher geometries F_1, F_2 are observably equivalent if

$$d_{\mathrm{obs}}(F_1, F_2) < \tau,$$

where d_{obs} is the battery-defined observable distance and τ is the observational threshold.

Observable equivalence underlies quotient Čencov stability.

2.5 Admissible Morphism Hierarchy

We distinguish three nested morphism classes:

$$\mathcal{M}_{\mathrm{Markov}} \supseteq \mathcal{M}_{\mathrm{Fisher}} \supseteq \mathcal{M}_{\mathrm{forced}}.$$

- $\mathcal{M}_{\mathrm{Markov}}$: all Markov morphisms acting on observables.
- $\mathcal{M}_{\mathrm{Fisher}}$: morphisms preserving Fisher distinguishability up to admissibility tolerance.
- $\mathcal{M}_{\mathrm{forced}}$: morphisms satisfying Fisher admissibility, inversion stability, and dimensional closure.

This hierarchy clarifies that CTMT does not introduce arbitrary morphisms; it restricts admissibility by intersecting independently justified constraints.

3 Recoverability Bounds and Approximate Null Preservation

We now connect inversion-stability to geometric properties of the forward operator, in particular to the image and null space of J.

Let $\Pi_{\mathrm{Im}(J)}$ denote the orthogonal projector onto $\mathrm{Im}(J)$.

Lemma 1 (Recoverability bound)**.** *If P is inversion-stable, then there exists $C(J, \lambda) > 0$ such that*

$$\|(I - \Pi_{\mathrm{Im}(J)})Py\| \le C(J, \lambda)\, \varepsilon_{\mathrm{inv}}$$

for all admissible y.

Lemma 2 (Approximate kernel stability)**.** *Let v be a right singular vector of J with singular value $\sigma_i > \sigma_{\mathrm{tol}}$. If P is inversion-stable, then*

$$\|PJv\| \leq C(J, \lambda)\, \varepsilon_{\mathrm{inv}}.$$

These lemmas replace exact equalities such as $\ker(J) = \ker(PJ)$ with stability bounds on identifiable and null-supported directions.

Proof 5. If $v \in \ker(J)$, then $Jv = 0$ and hence $J_\lambda^\dagger Jv = 0$. Inversion-stability implies that $J_\lambda^\dagger PJv$ remains small in norm. Since $J_\lambda^\dagger$ is bounded, this yields a bound on $\|PJv\|$ with a constant depending on J and λ.

Remark 2. Lemmas **??** and **??** do not assert exact equality of images or null spaces. Instead, they provide stability bounds: the image of J is approximately preserved under P, and null-supported directions remain approximately null. This is the appropriate level of generality for ill-posed or regularized problems and is much more defensible than exact equalities.

4 Dimensional closure and physical admissibility

In addition to Fisher admissibility and inversion-stability, CTMT applications often require dimensional closure: conservation of mass, charge, probability, or other physically meaningful quantities, as well as positivity and unit consistency.

Definition 6 (Dimensional closure)**.** A morphism P satisfies dimensional closure if it preserves the relevant conserved quantities and respects positivity and unit consistency. Formally, if L is a linear functional encoding a conserved quantity (e.g., total mass), then

$$L(Py) = L(y)$$

for all admissible y, and if y is constrained to a positive cone or a dimensionally consistent subspace, then so is Py.

We do not attempt to formalize all possible dimensional constraints here. The key point is that dimensional closure provides an independent admissibility filter that is not derived from Fisher or inversion alone.

5 Forced admissibility class

Definition 7. The forced admissibility class is

$$\mathcal{M}_{\mathrm{forced}} = \mathcal{M}_{\mathrm{Fisher}} \cap \mathcal{M}_{\mathrm{Inv}} \cap \mathcal{M}_{\mathrm{Dim}}.$$

Theorem 1 (Forced admissibility)**.** *If P satisfies Fisher admissibility, inversion stability, and dimensional closure, then:*

1. *distinguishability is preserved up to Fisher tolerance,*
2. *recoverability is preserved up to inversion tolerance,*
3. *null-supported directions remain approximately null,*
4. *resolved directions remain approximately identifiable.*

6 Connection to the synthetic coil battery

We now connect the forced admissibility class to the previously studied synthetic coil battery. In that work, a common local-anchor coil platform was used, with a fixed Jacobian/Fisher pipeline and a suite of attacked scenarios including baseline, current scaling, radius scaling, compensated attack, and explicit violation cases.

The coil battery reported, for each scenario, a structural transport signature

$$\Sigma_{\mathrm{tr}}(s) = \big(\mathbf{1}_{\mathrm{Null}}(s),\ \mathbf{1}_{\mathrm{Wind}}(s),\ \mathbf{1}_{\mathrm{Self}}(s),\ \mathbf{1}_{\mathrm{Cenc}}(s)\big),$$

where the binary entries indicated null-curvature stability, winding invariance, self-reference stability, and quotient-style Čencov stability. It was observed that baseline and current-scaling scenarios had

$$\Sigma_{\mathrm{tr}} = (1, 1, 1, 1),$$

while radius-scaling and compensated-attack scenarios had

$$\Sigma_{\mathrm{tr}} = (0, 1, 1, 0),$$

and violation scenarios failed all indicators.

Proposition 1 (Observed invariant selectivity on the coil battery)**.** *Within the synthetic coil battery:*

- *all gauge-candidate scenarios (baseline and current scaling) satisfied Fisher admissibility, null-curvature stability, winding invariance, self-reference stability, and quotient-style Čencov stability;*
- *dangerous compensated scenarios (radius scaling and compensated attack) satisfied Fisher admissibility, winding invariance, and self-reference stability, but failed null-curvature and quotient-style Čencov stability;*
- *violation scenarios failed Fisher admissibility and all structural indicators.*

Remark 3. This pattern shows that Fisher admissibility alone is not sufficient to decide gauge equivalence on the tested suite. The decisive non-gauge signature lies in null-curvature breakdown and quotient-style Čencov instability, whereas patched Fisher monotonicity, winding, and self-reference contraction can still survive. This is consistent with the present forced-class framework, where Fisher admissibility is only one of several admissibility layers.

Theorem 2 (Tested-class necessity)**.** *Within the synthetic coil family, every scenario satisfying Fisher admissibility, inversion stability, and dimensional closure preserved*

$$\Sigma_{\mathrm{tr}} = (1, 1, 1, 1).$$

Every violating scenario broke at least one component. Thus the structural signature is a necessary discriminator for the tested class.

This is a tested-class statement, not a universal law.

Proof 8. On the coil battery, the only scenarios that preserved all structural indicators were baseline and current scaling, which also satisfied Fisher admissibility and were not constructed to violate inversion or dimensional closure. Dangerous compensated scenarios failed null-curvature and quotient-style Čencov stability and were therefore structurally refused. Violation scenarios failed Fisher admissibility and structural indicators. Thus, within the tested class, the structural signature is necessary for survival under the combined admissibility filters.

Remark 4. Theorem 2 is explicitly a tested-class statement. It does not claim that the structural signature is a universal discriminator across all possible forward models, kernels, or physical systems. It claims that, within the synthetic coil family and under the stated admissibility conditions, the structural signature is necessary for survival. This is the strongest claim currently supported by the available evidence.

7 Synthetic Coil Validation Battery

We summarize the coil platform:

$$R = 5 \text{ cm}, \qquad n_z = 60, \quad n_r = 40, \qquad \sigma_B = 2 \times 10^{-8} \text{ T}.$$

Anchor:

$$\theta_{\text{anchor}} = (4.0,\ 5.8 \times 10^{-4},\ -4.2 \times 10^{-4}).$$

Singular values:

$$s(J) = (1.63,\ 1.74 \times 10^{-2},\ 1.64 \times 10^{-4}).$$

The battery tested baseline, current scaling, radius scaling, compensated attack, and violation scenarios. Only baseline and current scaling preserved the full structural signature

$$\Sigma_{\text{tr}} = (1, 1, 1, 1).$$

Radius scaling and compensated attack preserved Fisher monotonicity, winding, and self-reference but failed null-curvature and quotient Čencov stability.

8 Acceptance rules and falsifiability

We summarize the acceptance rules implied by the forced admissibility class and the coil battery.

Definition 9 (Acceptance rules)**.** A scenario is accepted as admissible on the tested class if:

(R1) Fisher admissibility holds within tolerance,

(R2) inversion stability holds within tolerance,

(R3) dimensional closure holds,

(R4) the structural transport signature satisfies

$$\Sigma_{\text{tr}} = (1, 1, 1, 1).$$

A scenario is refused if any of these conditions fails.

Proposition 2 (Tested-class falsifiability)**.** *The present framework is falsifiable in the following sense: if future synthetic or laboratory suites produce scenarios that satisfy Fisher admissibility, inversion stability, and dimensional closure but systematically fail one or more components of the structural signature while remaining physically gauge-equivalent, then the present structural necessity claim must be weakened, restricted, or abandoned for that class.*

Proof 10. If such scenarios exist, then the structural signature is no longer a necessary discriminator even on the tested class. The acceptance rules would then be too strict and would need to be revised. This is a standard falsifiability condition.

9 Limitations

Theorem 3 (Non-uniqueness outside the admissible class)**.** *The forced admissibility class* $\mathcal{M}_{\text{forced}}$ *is not claimed to be the unique physically admissible morphism class. It is the class forced by the simultaneous requirements of Fisher admissibility, inversion stability, and dimensional closure within the tested regime.*

This prevents overclaiming and clarifies the scope of the results.

10 Conclusion

We have constructed a forced admissibility class for CTMT transport morphisms by intersecting Fisher admissibility (Čencov), inversion stability (recoverability), and dimensional closure (physical consistency). Within this framework, structural invariants such as null-curvature stability and quotient-style Čencov stability are not arbitrary decorations but necessary consequences of simultaneous distinguishability and recoverability preservation, at least within the tested synthetic coil family.

The strongest claim made here is deliberately modest: on the tested class, the structural transport signature is a necessary discriminator for admissible scenarios. No universality beyond the tested class is asserted. The framework is explicitly falsifiable and can be weakened or restricted if future suites contradict the observed pattern.

This is a more conservative but mathematically stronger position than earlier CTMT drafts. It does not attempt to derive physics from CTMT. It positions CTMT as a coherent admissibility and structural-filtering machinery whose invariants survive nontrivial synthetic adversarial attacks and whose morphism class is constrained by independent necessity principles rather than arbitrary choice.

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